ASSIGNMENT - 05

Cohen-Sutherland Line Clipping Algorithm

Roll Number: 208 | a = 0 | b = 8

Personal input line: L6 = (40 + a, 20) to (190 - b, 180) = (40, 20) to (182, 180).

Clipping window: Lower-left = (50, 50), upper-right = (200, 150).

Region-code bit order: TOP, BOTTOM, RIGHT, LEFT.

Result Table

L1	(140, 70)	(185, 130)	0000	0000	Accepted	0	(140.000, 70.000)	(185.000, 130.000)
L2	(10, 165)	(40, 190)	1001	1001	Rejected	0	-	-
L3	(20, 125)	(250, 125)	0001	0010	Clipped	2	(50.000, 125.000)	(200.000, 125.000)
L4	(80, 20)	(80, 180)	0100	1000	Clipped	2	(80.000, 50.000)	(80.000, 150.000)
L5	(20, 20)	(150, 200)	0101	1000	Clipped	3	(50.000, 61.538)	(113.889, 150.000)
L6	(40, 20)	(182, 180)	0101	1000	Clipped	2	(66.625, 50.000)	(155.375, 150.000)

Figure 1 - Initial Diagram

All six original input lines are shown as dashed lines. The surviving portions are overlaid as solid lines.

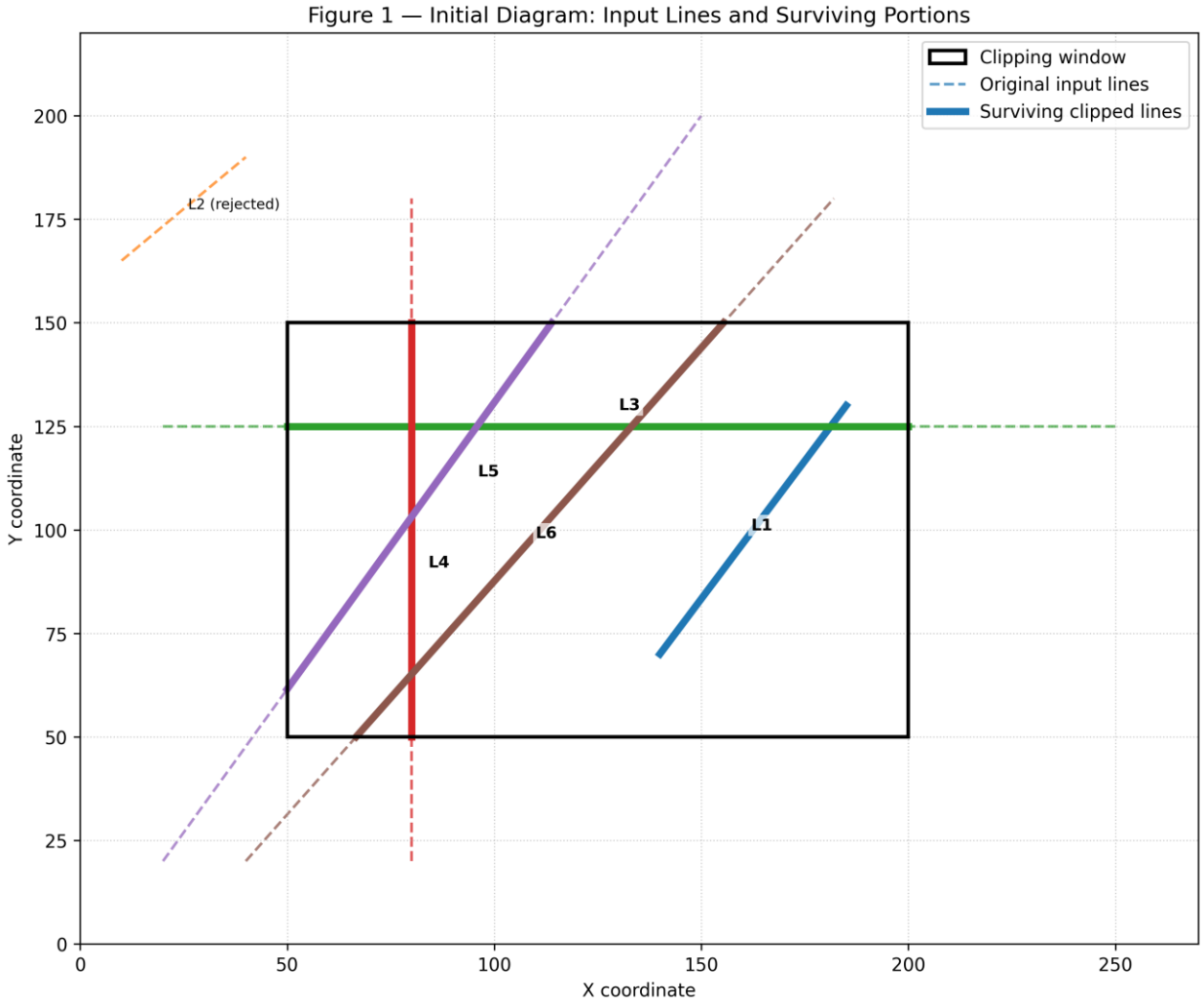
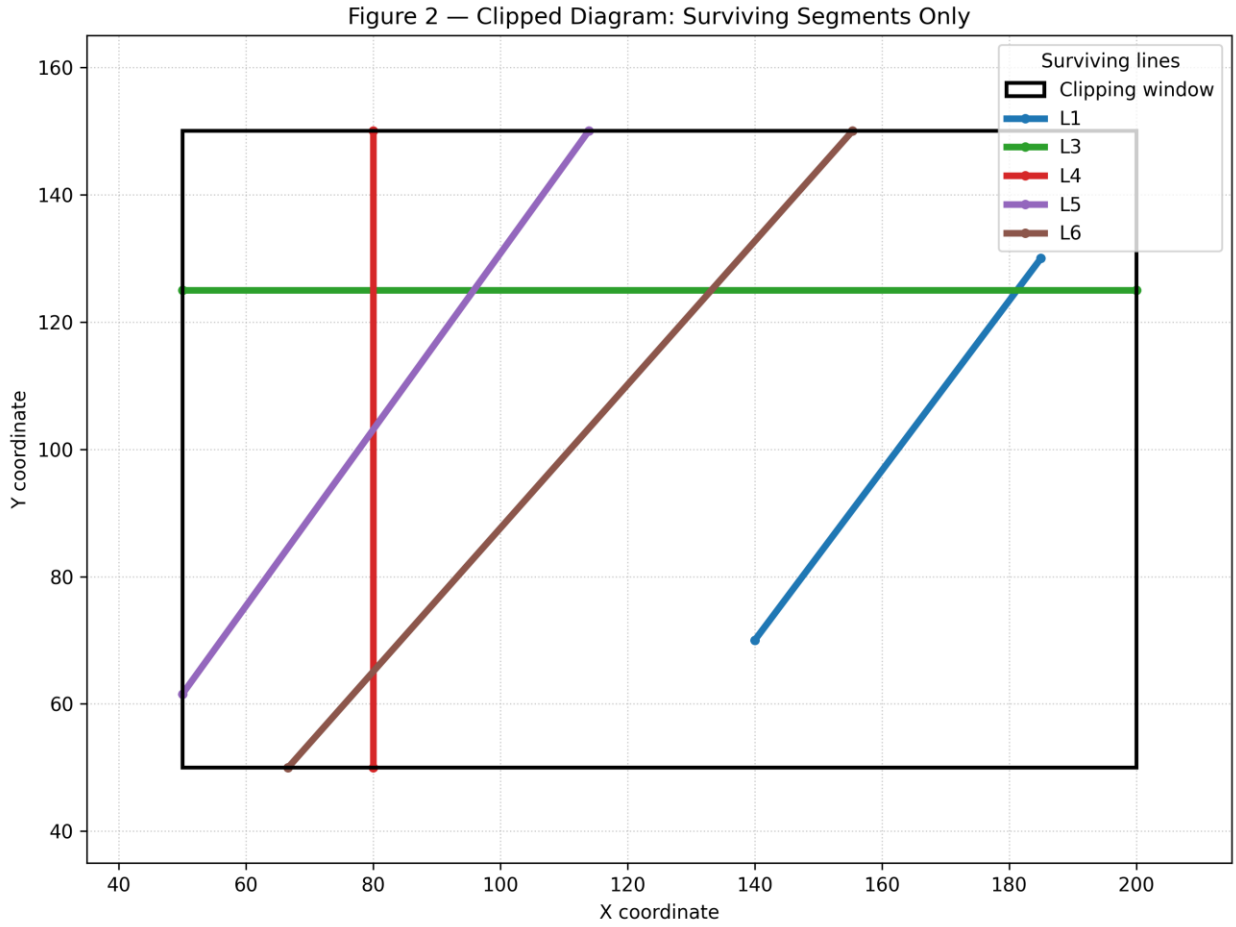


Figure 2 - Clipped Diagram

Only the portions that remain inside the clipping window are displayed. The rejected line L2 and all outside portions have been removed.



Questions to Answer

1Q. Which line was trivially accepted and which was trivially rejected? Give the region codes that told you, and state the bitwise test you used in each case.

A. L1 was trivially accepted because both endpoints have region code 0000. Their bitwise OR is 0000, satisfying the trivial acceptance condition.
L2 was trivially rejected because both endpoints have region code 1001. Their bitwise AND is 1001 (non-zero), satisfying the trivial rejection condition.

Bitwise tests:

Trivial acceptance: Code1 OR Code2 == 0000
Trivial rejection: Code1 AND Code2 != 0000

2Q. One of the six lines needed three passes through the loop, while the others needed two or none. Which line was it, and why did computing the first intersection point not finish the job?

A. L5 required three passes through the clipping loop. Its first intersection with the bottom boundary \(y=50\) was \((41.667, 50.000)\), which was still outside the window because its x-coordinate was less than 50. Therefore, another intersection with the left boundary \(x=50\) had to be calculated before clipping the remaining portion against the top boundary.

3Q. For your own line L6, which two edges of the window did it cross, and what are the clipped endpoints? Give the coordinates to three decimal places.

A. For roll number 208, my L6 line runs from \((40,20)\) to \((182,180)\). It crosses the bottom edge \((y=50)\) and the top edge \((y=150)\) of the clipping window.

The clipped endpoints, rounded to three decimal places, are:

- Bottom intersection: (66.625, 50.000)
- Top intersection: (155.375, 150.000)

Calculation :

For my roll number 208:

Starting point: P1 = (40, 20)

- Ending point: P2 = (182,180)
- Clipping window: 50 less than qual to x less than equal to 200,
  50 less than equal to y less than equal to 150

Since the line enters through the bottom boundary and exits through the top boundary, we calculate two intersections.

Now calculating intersection with the bottom edge

The bottom bounday is y = 50

Calculating the slop :

M = y2 – y1/ x2 – x1

M = 160/142

Now using the line quation : y – y1 = m(x – x1)

Now substituing the values will give us x = 66.625

Therefore, the first interaction is : p’1 = (66.625, 50.000)

Now Calculate intersection with the top edge

The bottom boundary is y = 150

Using the same line equation

We get x = 155.375

Hence the second intersection is p’2 = (155.375, 150.000)

4Q. The region code has four bits. How many different regions does this divide the plane into, and why is the code of a point inside the window 0000?

A. A four-bit region code can represent 16 different regions, because each bit can have two possible values (0 or 1).

$$2^4 = 16$$

A point inside the clipping window has the region code 0000 because it is neither above, below, or right, nor left of the window. Therefore, all four bits remain 0, indicating that the point lies inside the clipping boundary.